PROGRAM 1

```
import java.util.*;
class Difference{
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a number :"); int
num = sc.nextInt();
int rev = 0,dup = num;
while(dup > 0){
int d = dup % 10;
rev = rev*10 + d;
dup = dup/10; }
int dif = Math.abs(num - rev);
System.out.println("Reversed number : "+rev);
System.out.println("Absolute difference is " +dif);
}
}
```

OUTPUT

Output - Run (Difference)

```
Compiling 1 source file with javac [debug release 25] to target\class

--- exec:3.5.1:exec (default-cli) @ Difference ---
Enter a number :
 33
Reversed number : 33
Absolute difference is 0
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  9.686 s
Finished at: 2026-01-13T13:23:50+05:30
------------------------------------------------------------------------
```

PROGRAM 2

```
import java.util.*;
class Frequency{
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a word:");
String s = sc.next();
int l = s.length();
int c = 0;
for(int i = 0; i < l; i++){
char ch = s.charAt(i);
if("AEIOUaeiou".indexOf(ch) != -1){
System.out.print(ch);
c = c+1;
}
}
System.out.println();
System.out.println("The frequency of vowels = "+ c);
}
}
```

OUTPUT

Output - Run (Frequency)

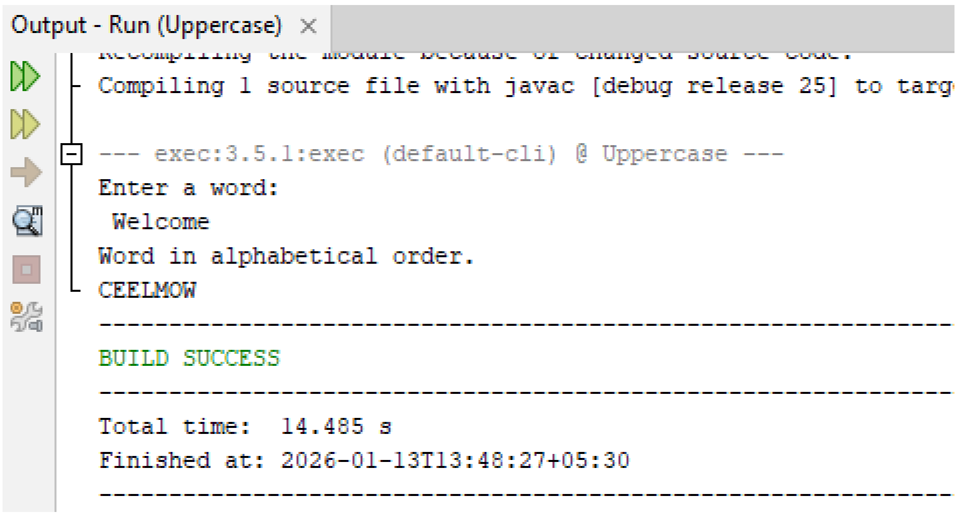
```
Compiling 1 source file with javac [debug release 25] to targe

--- exec:3.5.1:exec (default-cli) @ Frequency ---
Enter a word:
 Riya
ia
The frequency of vowels = 2
------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------
Total time:  06:34 min
Finished at: 2026-01-13T13:42:40+05:30
------------------------------------------------------------------
```

PROGRAM 3

```
import java.util.*;
class Uppercase {
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a word:");
String s = sc.next();
s = s.toUpperCase();
int l = s.length();
char ch;
System.out.println("Word in alphabetical order.");
for(int i = 65; i <= 90; i++){
for(int j = 0; j < l; j++){
ch = s.charAt(j);
if(ch == (char)i){
System.out.print(ch);
}
}
}
}
}
```

}

OUTPUT



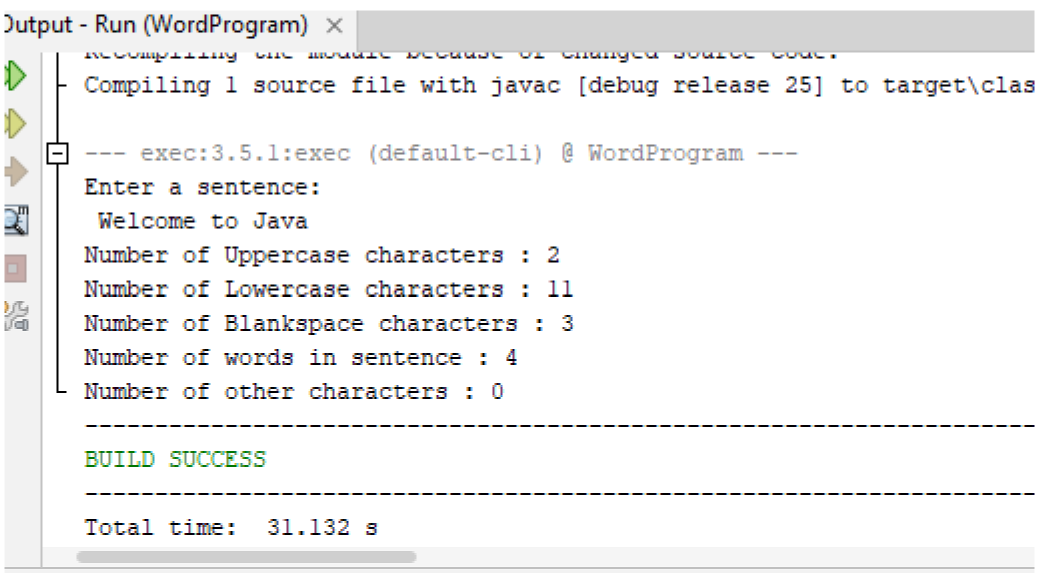
```
Output - Run (Uppercase)
Compiling 1 source file with javac [debug release 25] to targ

--- exec:3.5.1:exec (default-cli) @ Uppercase ---
Enter a word:
 Welcome
Word in alphabetical order.
CEELMOW
------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------
Total time:  14.485 s
Finished at: 2026-01-13T13:48:27+05:30
------------------------------------------------------------
```

PROGRAM 4

```
import java.util.*;
class WordProgram {
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a sentence:");
String s = sc.nextLine();
int u = 0,l = 0,b = 0,w = 1,o = 0; int
len = s.length();
for(int i = 0; i < len; i++){
char ch = s.charAt(i);
if(Character.isUpperCase(ch)){
u = u+1;
}else if(Character.isLowerCase(ch)){
l = l+1;
}else if(Character.isWhitespace(ch)){
b = b+1;
w = w+1;
```

```
}else{
o = o+1;
}
}
System.out.println("Number of Uppercase characters : "+u);
System.out.println("Number of Lowercase characters : "+l);
System.out.println("Number of Blankspace characters : "+b);
System.out.println("Number of words in sentence : "+w);
System.out.println("Number of other characters : "+o);
}
}
```

OUTPUT

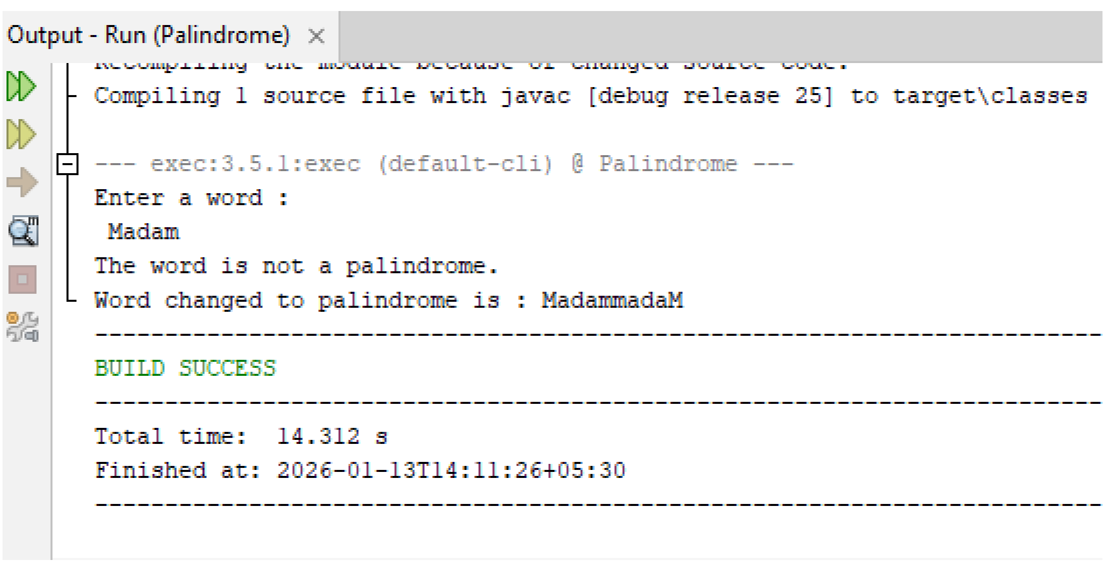


PROGRAM 5

```
import java.util.*;
class Palindrome {
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a word :");
```

```
String s = sc.next();
char ch ;
String s1 = "";
int l =s.length();
for(int i =0; i < l; i++){
ch = s.charAt(i);
s1 = ch + s1;
}
if(s.equals(s1)){
System.out.println("The word is a palindrome.");
}else{
System.out.println("The word is not a palindrome.");
String s2 = s + s1;
System.out.println("Word changed to palindrome is : "+ s2);
}
}
}
```

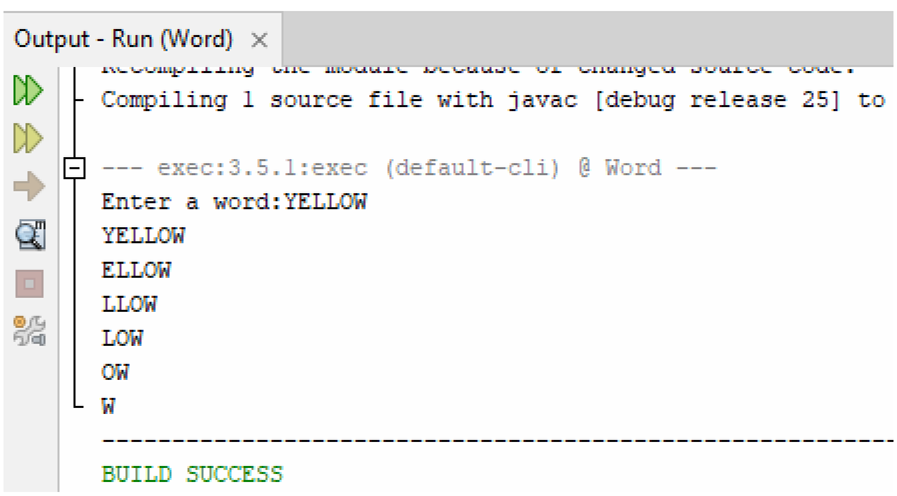
OUTPUT



PROGRAM 6 :

```
import java.util.Scanner;
public class Word
{
public static void main(String args[]) {
Scanner in = new Scanner(System.in);
System.out.print("Enter a word:");
String word = in.nextLine();
int len = word.length();
for (int i = 0; i<len; i++) {
for (int j = i; j<len; j++) {
char ch = word.charAt(j);
System.out.print(ch); }
System.out.println();
}
}
}
```

OUTPUT:

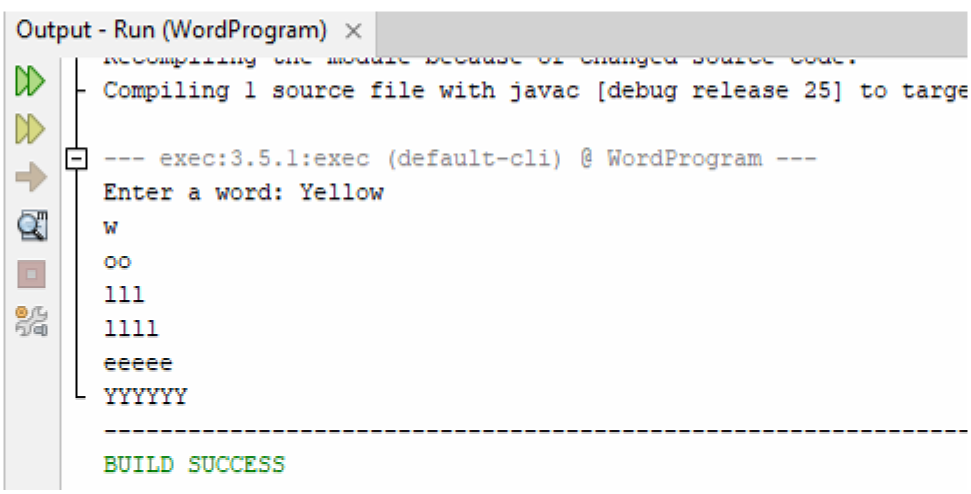


PROGRAM 7:

```
import java.util.Scanner;
```

```
public class WordProgram
{
public static void main(String args[])
{
Scanner in = new Scanner(System.in);
System.out.print("Enter a word: ");
String word = in.nextLine();
int len = word.length();
for (int i = len - 1; i>= 0; i--)
{
for (int j = len - 1; j>= i; j--)
{
char ch = word.charAt(i);
System.out.print(ch);
}
System.out.println();
}
}
}
```

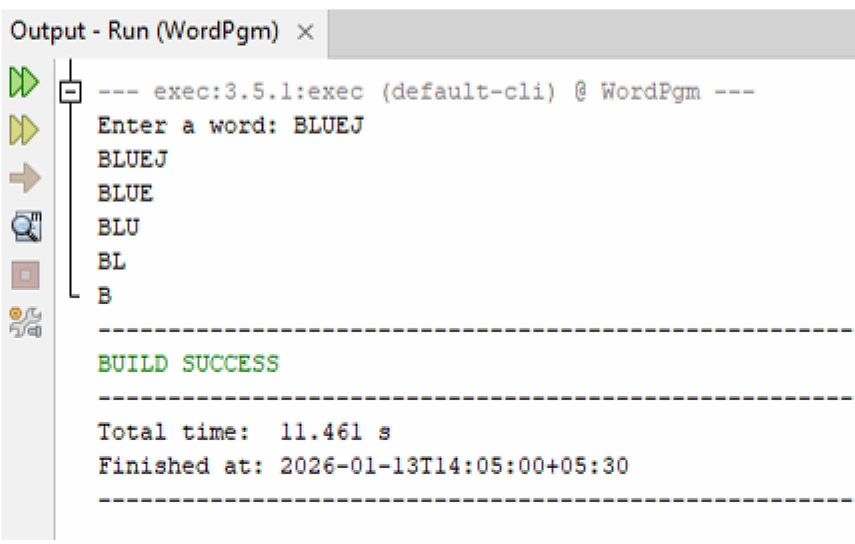
OUTPUT:



PROGRAM 8:

```java
import java.util.Scanner;
public class WordPgm {
public static void main(String args[]) {
Scanner in = new Scanner(System.in);
System.out.print("Enter a word: ");
String word = in.nextLine();
int len = word.length();
for (int i = len; i > 0; i--) {
for (int j = 0; j < i; j++) {
System.out.print(word.charAt(j));
}
System.out.println();
}
}
}
```

OUTPUT:



PROGRAM 9:

```java
import java.util.Scanner;
public class KboatSDANames
{
```

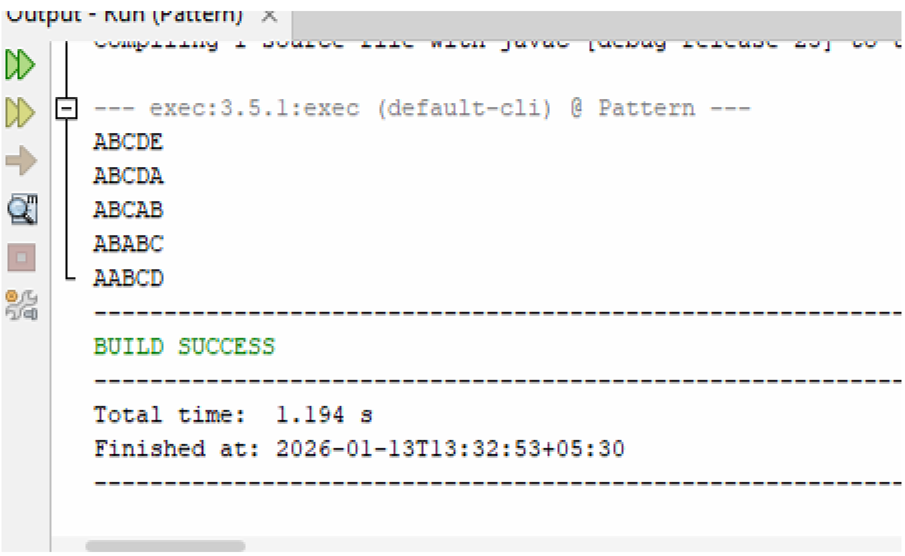
```
public static void main(String args[]) {
Scanner in = new Scanner(System.in);
String names[] = new String[10];
System.out.println("Enter 10 names");
for (int i = 0; i<names.length; i++) {
names[i] = in.nextLine();
}
System.out.print("Enter a letter:");
char ch = in.next().charAt(0);
ch = Character.toUpperCase(ch);
for (int i = 0; i<names.length; i++) {
if (Character.toUpperCase(names[i].charAt(0)) == ch) {
System.out.println(names[i]);
}
}
}
}
```

OUTPUT:

PROGRAM 10:

```
public class Pattern
{
public static void main(String args[]) {
String word = "ABCDE";
int len = word.length();
for (int i = 0; i<len; i++) {
for (int j = 0; j<len - i; j++) {
System.out.print(word.charAt(j));
}
for (int k = 0; k<i; k++) {
System.out.print(word.charAt(k));
}
System.out.println();
}
}
}
```

OUTPUT :

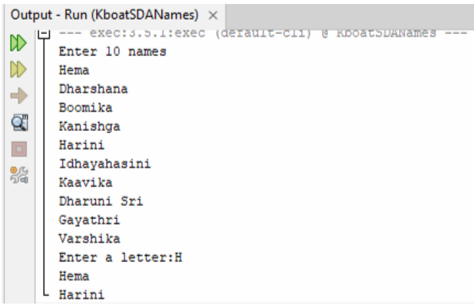


PROGRAM 11:

```
import java.util.Scanner;
public class KboatSDANames
{
public static void main(String args[]) {
Scanner in = new Scanner(System.in);
String names[] = new String[10];
System.out.println("Enter 10 names");
for (int i = 0; i<names.length; i++) {
names[i] = in.nextLine();
}
System.out.print("Enter a letter:");
char ch = in.next().charAt(0);
ch = Character.toUpperCase(ch);
for (int i = 0; i<names.length; i++) {
if (Character.toUpperCase(names[i].charAt(0)) == ch) {
System.out.println(names[i]);
}
}
}
}
```

OUTPUT:

```
}
```



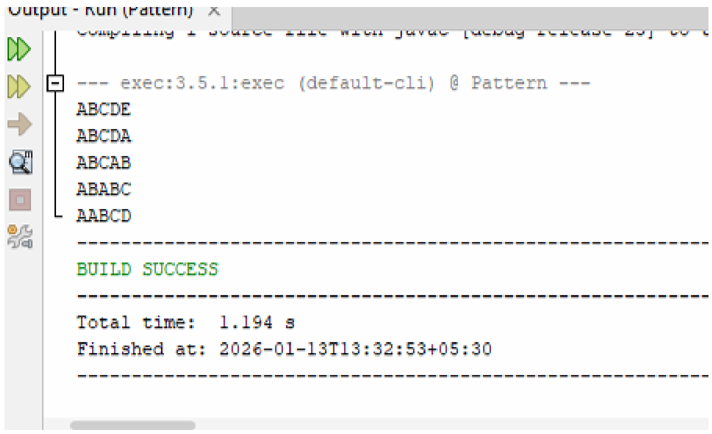
```
--- exec:3.5.1:exec (default-cli) @ KboatSDANames ---
Enter 10 names
Hema
Dharshana
Boomika
Kanishga
Harini
Idhayahasini
Kaavika
Dharuni Sri
Gayathri
Varshika
Enter a letter:H
Hema
Harini
```

PROGRAM 12:

```
public class Pattern

public class Pattern{

public static void main(String args[]) {

String word = "ABCDE";

int len = word.length();

for (int i = 0; i<len; i++) {

for (int j = 0; j<len - i; j++) {

System.out.print(word.charAt(j));

}

for (int k = 0; k<i; k++) {

System.out.print(word.charAt(k));

}

System.out.println();

}

}
```

}

OUTPUT :

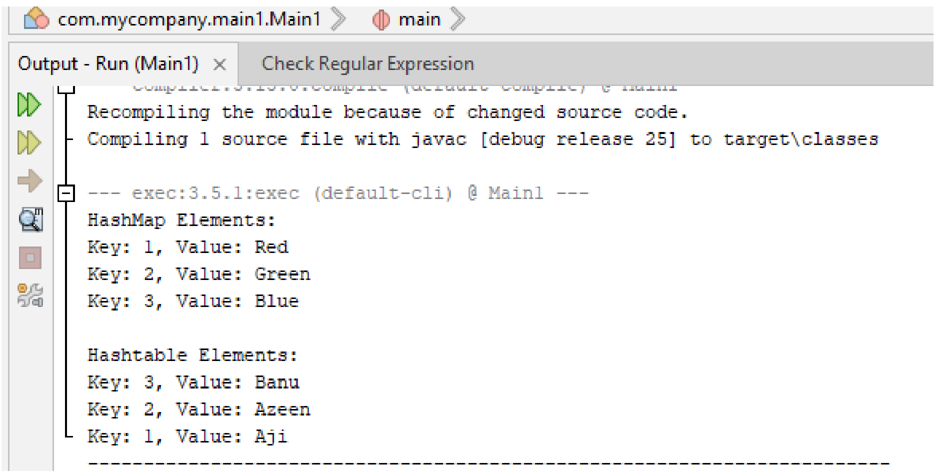


PROGRAM 13:

```
import java.util.*;
public class Main1 {
public static void main(String[] args) {
HashMap<Integer, String> hashMap = new HashMap<>();
hashMap.put(1, "Red");
hashMap.put(2, "Green");
hashMap.put(3, "Blue");
System.out.println("HashMap Elements: ");
for (Integer key : hashMap.keySet()) {
System.out.println("Key: " + key + ", Value: " + hashMap.get(key));
}
Hashtable<Integer, String> hashtable = new Hashtable<>();
hashtable.put(1, "Aji");
hashtable.put(2, "Azeen");
hashtable.put(3, "Banu");
System.out.println("\nHashtable Elements:");
```

```java
for (Integer key : hashtable.keySet()) {
System.out.println("Key: " + key + ", Value: " + hashtable.get(key));
}
}
}
```

OUTPUT

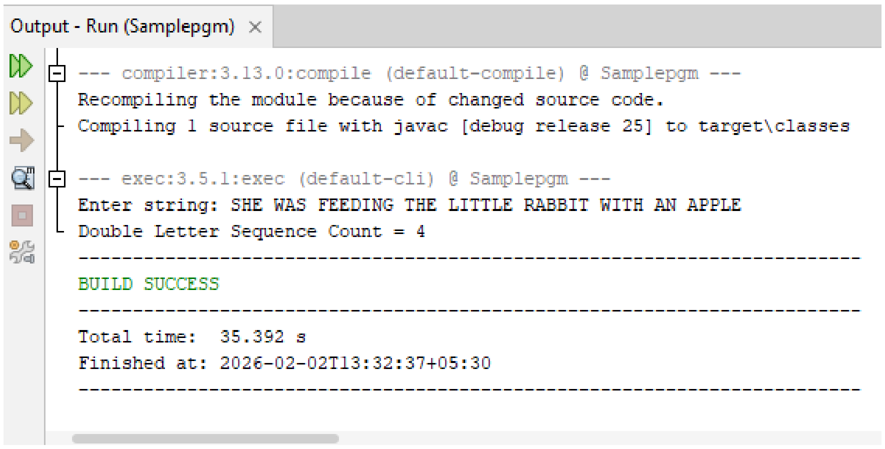


PROGRAM 14:

```java
import java.util.Scanner;
public class Samplepgm {
public static void main(String args[]) {
Scanner in = new Scanner(System.in);
System.out.print("Enter string: ");
String s = in.nextLine();
String str = s.toUpperCase(); int
count = 0;
int len = str.length();
for (int i = 0; i < len - 1; i++) {
if (str.charAt(i) == str.charAt(i + 1))
count++;
}
```

System.out.println("Double Letter Sequence Count = " + count);
}
}

OUTPUT:



Output - Run (Samplepgm)

```
--- compiler:3.13.0:compile (default-compile) @ Samplepgm ---
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to target\classes

--- exec:3.5.1:exec (default-cli) @ Samplepgm ---
Enter string: SHE WAS FEEDING THE LITTLE RABBIT WITH AN APPLE
Double Letter Sequence Count = 4
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  35.392 s
Finished at: 2026-02-02T13:32:37+05:30
------------------------------------------------------------------------
```